

Solutions

Numbering follows the lecture notes. For TYPE C exercises the idea is boxed; the rest are routine once the idea is in hand, which is what the types mean.

Lectures 3 and 4

§1.2 What “converging to a Gaussian” means

1.3 C

The hypothesis already bounds how much mass X_n puts far out: test it against a **smooth cutoff**. Tightness is free, and after that only a mollification is left.

Let χ be smooth with $0 \leq \chi \leq 1$, $\chi = 1$ on $\{|x| \leq 1\}$ and $\chi = 0$ on $\{|x| \geq 2\}$, and put $\chi_M(x) = \chi(x/M)$, which is in the test class for each M . Since $\mathbf{1}\{|X| > M\} \rightarrow 0$ pointwise as $M \rightarrow \infty$ and is bounded by 1, dominated convergence gives $\mathbb{P}(|X| > M) \rightarrow 0$; choose M with $\mathbb{P}(|X| > M) < \varepsilon$. Then $\mathbb{E}\chi_M(X) \geq \mathbb{P}(|X| \leq M) > 1 - \varepsilon$, so

$$\limsup_n \mathbb{E}[1 - \chi_M(X_n)] = 1 - \mathbb{E}\chi_M(X) < \varepsilon.$$

Now let f be bounded continuous and set $g = f\chi_M$, continuous with support in $\{|x| \leq 2M\}$ and therefore uniformly continuous. For $g \sim \mathcal{N}(0, I_k)$ put $g_\delta(x) = \mathbb{E}[g(x + \delta g)] = \int g(y)\phi_\delta(x - y)dy$. Differentiating under the integral — the difference quotients of ϕ_δ are dominated by an integrable function of y — each derivative $\partial^\alpha g_\delta(x) = \int g(y)\partial^\alpha \phi_\delta(x - y)dy$ is bounded by $\|g\|_\infty \|\partial^\alpha \phi_\delta\|_{L^1}$, so g_δ lies in the test class; and with ω the modulus of continuity of g ,

$$|g_\delta(x) - g(x)| \leq \mathbb{E}|g(x + \delta g) - g(x)| \leq \mathbb{E}[\min(\omega(\delta|g|), 2\|g\|_\infty)] \xrightarrow{\delta \rightarrow 0} 0$$

uniformly in x . Finally

$$|\mathbb{E}f(X_n) - \mathbb{E}f(X)| \leq \|f\|_\infty \mathbb{E}[1 - \chi_M(X_n)] + \|f\|_\infty \mathbb{E}[1 - \chi_M(X)] + 2\|g - g_\delta\|_\infty + |\mathbb{E}g_\delta(X_n) - \mathbb{E}g_\delta(X)|.$$

Take $\limsup_n$: the last term vanishes by hypothesis and the first two are at most $2\|f\|_\infty \varepsilon$. Let $\delta \rightarrow 0$, then $\varepsilon \rightarrow 0$.

§1.6 Lindeberg and Feller

1.13 A Here $s_n^2 = n\sigma^2$ and every row entry has the law of X_1 , so

$$L_n(\varepsilon) = \frac{1}{n\sigma^2} \sum_{i=1}^n \mathbb{E}[X_i^2 \mathbf{1}\{|X_i| > \varepsilon\sigma\sqrt{n}\}] = \frac{1}{\sigma^2} \mathbb{E}[X_1^2 \mathbf{1}\{|X_1| > \varepsilon\sigma\sqrt{n}\}].$$

The integrand tends to 0 pointwise and is dominated by X_1^2 , which is integrable; dominated convergence gives $L_n(\varepsilon) \rightarrow 0$. So Theorem 1.9 applies with two moments only, where Theorem 1.7 assumed three. The third moment is not wasted: it is what buys the rate $n^{-1/2}$, which Theorem 1.9 does not supply.

1.14 C

Feller’s condition is blind to the rare large value; Lindeberg’s sees only that. The array is a Gaussian and a Poisson superimposed, and the limit records both.

$\text{Var}(X_{n,i}) = 1 \cdot (1 - \frac{1}{n}) + n \cdot \frac{1}{n} = 2 - \frac{1}{n}$, so $s_n^2 = 2n - 1$ and

$$\max_i \frac{\text{Var}(X_{n,i})}{s_n^2} = \frac{2 - \frac{1}{n}}{2n - 1} \rightarrow 0 :$$

Feller’s condition holds. For $\varepsilon \leq 1/\sqrt{2}$ we have $\varepsilon^2 \leq \frac{1}{2} < \frac{n}{2n-1}$, i.e. $\varepsilon s_n < \sqrt{n}$, while $\varepsilon s_n > 1$ once n is large; so $\{|X_{n,i}| > \varepsilon s_n\} = \{|X_{n,i}| = \sqrt{n}\}$ and

$$L_n(\varepsilon) = \frac{1}{2n-1} \cdot n \cdot \left(n \cdot \frac{1}{n}\right) = \frac{n}{2n-1} \rightarrow \frac{1}{2}.$$

Lindeberg’s condition fails. Feller’s holds, so Remark 1.10(c) applies in contrapositive form: the array does *not* satisfy the central limit theorem. This is the array promised in Remark 1.10(b).

Directly: $\varphi_{n,i}(t) = (1 - \frac{1}{n}) \cos t + \frac{1}{n} \cos(t\sqrt{n})$, so with $s_n = \sqrt{2n-1}$,

$$\varphi_{T_n}(t) = \left[\left(1 - \frac{1}{n}\right) \cos \frac{t}{s_n} + \frac{1}{n} \cos \frac{t\sqrt{n}}{s_n} \right]^n = \left[1 - \frac{1}{n} \left(\frac{t^2}{4} + 1 - \cos \frac{t}{\sqrt{2}} \right) + o\left(\frac{1}{n}\right) \right]^n,$$

using $\cos(t/s_n) = 1 - \frac{t^2}{2(2n-1)} + O(n^{-2})$ and $t\sqrt{n}/s_n \rightarrow t/\sqrt{2}$. Hence

$$\varphi_{T_n}(t) \rightarrow \exp\left(-\frac{t^2}{4}\right) \cdot \exp\left(\cos \frac{t}{\sqrt{2}} - 1\right),$$

which is $\mathcal{N}(0, \frac{1}{2})$ convolved with a compound Poisson of rate 1 and jumps $\pm 1/\sqrt{2}$. Half the variance came from the ± 1 's and became Gaussian; the other half came from the rare $\pm\sqrt{n}$'s and did not. At $t = \sqrt{2}\pi$ the limit is $e^{-\pi^2/2-2}$ against $e^{-\pi^2/2}$ for the standard Gaussian.

§1.7 Records

1.21 A The array $X_{n,i} = B_{n,i} - p_{n,i}$ is independent and centered with $|X_{n,i}| \leq 1$ and $s_n = \sigma_n \rightarrow \infty$, so Lemma 1.15 gives Lindeberg's condition and Theorem 1.9 the conclusion. For $p_{n,i} = \lambda/n$ the total variance is $\sigma_n^2 = \lambda(1 - \lambda/n) \rightarrow \lambda$, which is bounded: the hypothesis $s_n \rightarrow \infty$ of Definition 1.8 fails, so nothing above applies, and the limit is $\text{Poisson}(\lambda)$ — Exercise 1.33, or Theorem 2.10 with a rate.

1.22 B

Weak convergence survives affine maps that converge to the identity, provided no mass escapes to infinity. A uniform second moment is what rules that out.

The tool. Let f be bounded and uniformly continuous with modulus of continuity ω_f , and put $\delta_n = |a_n - 1|M + |b_n|$. On $\{|Z_n| \leq M\}$ the two arguments differ by at most δ_n , so the integrands differ by at most $\omega_f(\delta_n)$; off that event bound the difference by $2\|f\|_\infty$ and use $\mathbb{P}(|Z_n| > M) \leq \mathbb{E}Z_n^2/M^2 \leq C/M^2$. Hence

$$|\mathbb{E}f(a_n Z_n + b_n) - \mathbb{E}f(Z_n)| \leq \omega_f(\delta_n) + \frac{2\|f\|_\infty C}{M^2}.$$

Let $n \rightarrow \infty$ with M fixed, then $M \rightarrow \infty$: the left side tends to 0, and $\mathbb{E}f(Z_n) \rightarrow \mathbb{E}f(Z)$ by hypothesis.

The application. With $Z_n = (R_n - H_n)/s_n$, which has $\mathbb{E}Z_n^2 = 1$,

$$\frac{R_n - \log n}{\sqrt{\log n}} = \frac{s_n}{\sqrt{\log n}} Z_n + \frac{H_n - \log n}{\sqrt{\log n}},$$

and $s_n^2/\log n = (H_n - \sum_{i \leq n} i^{-2})/\log n \rightarrow 1$ while $H_n - \log n \rightarrow \gamma$, so the numerator of the second term is bounded. Theorem 1.17 and the tool give the claim.

Why Lemma 1.2 is not enough. Lemma 1.2 upgrades convergence against a *fixed* family of test functions. Here the quantity to control is $\mathbb{E}f(a_n Z_n + b_n)$, which is $\mathbb{E}f_n(Z_n)$ for a test function f_n that moves with n ; convergence for each fixed f says nothing about it directly. The uniform second moment is what closes the gap.

1.23 B Each sequence of choices $(c_1, \dots, c_n)$ with $c_i \in \{1, \dots, i\}$ has probability $\prod_{i \leq n} \frac{1}{i} = \frac{1}{n!}$, and there are $n!$ of them. The map to permutations is injective — from the finished permutation, delete n from its cycle and read off what it did, then recurse — hence a bijection onto S_n , so the construction produces the uniform law. Element i opens a new cycle exactly when its choice is the distinguished one, which happens with probability $1/i$, and the choices are independent by construction. So K_n has the law of $\sum_{i \leq n} \text{Bernoulli}(1/i)$ with independent summands, which is the array of Theorem 1.17; Exercise 1.22 supplies the centring by $\log n$.

§1.8* A wide network at initialization

1.25 B $\mathbb{E}V_j = \mathbb{E}[a_j] \mathbb{E}[\sigma(\langle w_j, x_i \rangle)] = 0$: independence of a_j from w_j is what lets the expectation factor, and $\mathbb{E}a_j = 0$ kills it. Likewise $\text{Cov}(V_j)_{il} = \mathbb{E}[a_j^2] \mathbb{E}[\sigma(\langle w_j, x_i \rangle)\sigma(\langle w_j, x_l \rangle)] = K(x_i, x_l)$, the same factoring with $\mathbb{E}a_j^2 = \sigma_a^2$. The V_j are i.i.d. because the pairs (a_j, w_j) are, and $(f_m(x_1), \dots, f_m(x_k)) = m^{-1/2} \sum_j V_j$ by definition.

The multidimensional central limit theorem then applies with only a second moment. The second route is more elementary but assumes more. For fixed $\xi \in \mathbb{R}^k$, Theorem 1.7 applied to the scalars $\langle \xi, V_j \rangle$ gives $\langle \xi, m^{-1/2} \sum_j V_j \rangle \Rightarrow \mathcal{N}(0, \xi^\top K \xi)$, hence $\varphi_{m^{-1/2} \sum_j V_j}(\xi) = \mathbb{E}[e^{-i\langle \xi, \cdot \rangle}] \rightarrow e^{-\frac{1}{2}\xi^\top K \xi}$ for every ξ , and Lemma 1.29 finishes. But Theorem 1.7 is stated with $\mathbb{E}|X_1|^3 < \infty$, so this route needs a third moment.

1.26 A $\sum_{i,l} v_i v_l K(x_i, x_l) = \sigma_a^2 \mathbb{E}[(\sum_i v_i \sigma(\langle w, x_i \rangle))^2] \geq 0$. K is a covariance because it is one — of V_j — so no condition on σ beyond square-integrability could have been needed.

1.27 C

$(\langle w, x \rangle, \langle w, x' \rangle)$ is a centered Gaussian pair, and by Theorem 1.5 of Lectures 1–2 its law depends on x, x' only through the 2×2 Gram matrix. **The d -dimensional integral is two-dimensional.**

σ is positively homogeneous, so K scales as $\|x\|\|x'\|$ and we may take $\|x\| = \|x'\| = 1$. The pair $(\langle w, x \rangle, \langle w, x' \rangle)$ is centered Gaussian with covariance $\begin{pmatrix} 1 & \cos \theta \\ \cos \theta & 1 \end{pmatrix}$, hence equal in law to $(g_1, \cos \theta g_1 + \sin \theta g_2)$ for independent standard g_1, g_2 .

Write $(g_1, g_2) = R(\cos \alpha, \sin \alpha)$ with $R \geq 0$ and α uniform on $[0, 2\pi)$ independent, $\mathbb{E}R^2 = 2$. Then the pair is $(R \cos \alpha, R \cos(\alpha - \theta))$ and

$$\mathbb{E}[\sigma \sigma'] = \mathbb{E}[R^2] \cdot \frac{1}{2\pi} \int_0^{2\pi} (\cos \alpha)_+ (\cos(\alpha - \theta))_+ d\alpha = \frac{1}{\pi} \int_{\theta-\pi/2}^{\pi/2} \cos \alpha \cos(\alpha - \theta) d\alpha,$$

the integrand being positive exactly on that interval, of length $\pi - \theta$. Now $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$ gives

$$\int_{\theta-\pi/2}^{\pi/2} \cos \alpha \cos(\alpha - \theta) d\alpha = \frac{1}{2}(\pi - \theta) \cos \theta + \frac{1}{4} \left[\sin(2\alpha - \theta) \right]_{\theta-\pi/2}^{\pi/2} = \frac{1}{2}[(\pi - \theta) \cos \theta + \sin \theta],$$

since the bracket evaluates to $\sin \theta - (-\sin \theta) = 2 \sin \theta$. Hence the value is $\frac{1}{2\pi}(\sin \theta + (\pi - \theta) \cos \theta)$.

Checks. At $\theta = 0$: $\frac{1}{2\pi} \cdot \pi = \frac{1}{2} = \mathbb{E}[\max(0, g)^2]$. At $\theta = \pi/2$ the two projections are independent, so the answer should be $\mathbb{E}[\max(0, g)]^2 = (2\pi)^{-1}$, and it is.

§1.9* The proof we did not give

1.30 A The j th difference quotient of $t \mapsto e^{-itX}$ is dominated by $|X|^j$, integrable for $j \leq k$, so we may differentiate $\varphi_X(t) = \mathbb{E}[e^{-itX}]$ under the expectation k times, bringing down $(-iX)^k$. For $\varphi = e^{-t^2/2}$: $\varphi' = -t\varphi$, $\varphi'' = (t^2 - 1)\varphi$, so $\varphi''(0) = -1 = (-i)^2 \mathbb{E}g^2$. The series $e^{-t^2/2} = 1 - \frac{t^2}{2} + \frac{t^4}{8} - \dots$ has t^4 coefficient $\frac{1}{8}$, so $\varphi^{(4)}(0) = 4!/8 = 3 = (-i)^4 \mathbb{E}g^4$.

1.32 B With $a = 1 + z_n/n$ and $b = e^{z_n/n}$, both are $1 + O(1/n)$ so $|a|, |b| \leq M$ for some fixed M , and $|a - b| = O(1/n^2)$; hence $|a^n - b^n| \leq nM^{n-1}|a - b| \rightarrow 0$ while $b^n = e^{z_n} \rightarrow e^z$. Applying this with $z_n = n(\varphi_{X_1}(t/\sqrt{n}) - 1) \rightarrow -t^2/2$ by Lemma 1.31 gives $\varphi_{S_n}(t) \rightarrow e^{-t^2/2}$, and Lemma 1.29 finishes. **Independence** is what gives $\varphi_{S_n} = \prod_i \varphi_{X_i}(t/\sqrt{n})$; **identical distribution** only lets us write one factor n times.

1.33 A $\varphi_B(t) = (1 - p) + pe^{-it} = 1 + p(e^{-it} - 1)$. So $\varphi_{N_n} = [1 + \frac{\lambda}{n}(e^{-it} - 1)]^n \rightarrow \exp(\lambda(e^{-it} - 1))$ by Exercise 1.32, and $\sum_k e^{-\lambda} \frac{\lambda^k}{k!} e^{-itk} = e^{-\lambda} e^{\lambda e^{-it}}$ is the same thing.

1.34 B $X_{n,i}$ takes $1 - \lambda/n$ with probability λ/n and $-\lambda/n$ otherwise, so $s_n^2 = n \cdot \frac{\lambda}{n}(1 - \frac{\lambda}{n}) = \lambda(1 - \frac{\lambda}{n})$. For $\varepsilon < 1/\sqrt{\lambda}$ the threshold $\varepsilon s_n \rightarrow \varepsilon\sqrt{\lambda} < 1$ eventually falls below the large value, so $L_n(\varepsilon) \rightarrow n \cdot \frac{\lambda}{n} \cdot 1/\lambda = 1$.

At $\lambda = 3$, $\varepsilon = \frac{1}{2}$: $n = 10$ gives $s_n = 1.449$ and $\varepsilon s_n = 0.7246$, while the large value is $1 - \frac{3}{10} = 0.7 < 0.7246$ — nothing is caught, so $L_{10} = 0$ **exactly**. Then 0.970 and 1.000.

1.35 C

A Gaussian characteristic function $e^{-\sigma^2 t^2/2}$ **never vanishes**, and weak convergence forces the transforms to converge. So one zero rules out every Gaussian limit, with no need to know that a limit exists.

$s_n^2 = n \cdot 1 + n = 2n$. The large summand has $|X_{n,n+1}| = \sqrt{n} > \varepsilon\sqrt{2n} = \varepsilon s_n$ exactly when $\varepsilon < 1/\sqrt{2}$, and it alone contributes $n/2n = \frac{1}{2}$ to L_n . The first n summands give $s_n^{-1} \sum_{i \leq n} X_{n,i} \Rightarrow \mathcal{N}(0, \frac{1}{2})$ with transform $e^{-t^2/4}$; the last gives $\pm 1/\sqrt{2}$ with transform $\cos(t/\sqrt{2})$; they are independent, so $\varphi_n(t) \rightarrow e^{-t^2/4} \cos(t/\sqrt{2})$, which vanishes at $t = \pi/\sqrt{2}$. If $s_n^{-1} \sum_i X_{n,i} \Rightarrow \mathcal{N}(0, \sigma^2)$ then, $e^{-it \cdot}$ being bounded and continuous, $\varphi_n(t) \rightarrow e^{-\sigma^2 t^2/2}$ for every t ; at $t = \pi/\sqrt{2}$ that is impossible.

1.36 C

S_n has an **atom**. A distribution function with a jump of size a is at least $a/2$ from any continuous one.

If $F(z_0) - F(z_0^-) = a$ and Φ is continuous then $|F(z_0) - \Phi(z_0)| + |F(z_0^-) - \Phi(z_0^-)| \geq a$, so the sup is $\geq a/2$. For $X_i = \pm 1$ and n even, $\mathbb{P}(S_n = 0) = \binom{n}{n/2} 2^{-n} \sim \sqrt{2/\pi n}$, giving $\sup_z |\mathbb{P}(S_n \leq z) - \Phi(z)| \geq c n^{-1/2}$. Lattice summands were slowest in §1.6 for exactly this reason: their atoms cannot be smoothed, and they are of the same order as the whole error budget.

1.37 C

Compare $1 - \varphi_Y$ against $1 - \frac{\sin x}{x}$, which is non-negative everywhere and at least $\frac{1}{2}$ once $|x| \geq 2$.

Since $\varphi_Y(-t) = \overline{\varphi_Y(t)}$ the integral is real, and by Fubini — the integrand is bounded by 1 on a bounded interval — $\int_{-u}^u \varphi_Y(t) dt = \mathbb{E} \int_{-u}^u e^{-itY} dt = \mathbb{E} \left[\frac{2 \sin(uY)}{Y} \right]$, so

$$\frac{1}{u} \int_{-u}^u (1 - \varphi_Y(t)) dt = 2 \mathbb{E} \left[1 - \frac{\sin(uY)}{uY} \right].$$

The bracket is non-negative because $|\sin x| \leq |x|$, and at least $\frac{1}{2}$ when $|uY| \geq 2$ because $|\sin x/x| \leq 1/|x|$. Hence the right side is at least $2 \cdot \frac{1}{2} \mathbb{P}(|uY| \geq 2) = \mathbb{P}(|Y| \geq 2/u)$.

For tightness, $\psi(0) = \lim_n \varphi_{Y_n}(0) = 1$, so continuity of ψ at 0 gives, for any $\varepsilon > 0$, a $u > 0$ with $\frac{1}{u} \int_{-u}^u (1 - \psi) < \varepsilon/2$. The integrands are bounded by 2, so dominated convergence gives $\frac{1}{u} \int_{-u}^u (1 - \varphi_{Y_n}) \rightarrow \frac{1}{u} \int_{-u}^u (1 - \psi)$ and $\mathbb{P}(|Y_n| \geq 2/u) < \varepsilon$ for all $n \geq N$. Increasing K past $2/u$ covers the finitely many remaining n .

§2.1 The characterization

2.3 B

Both sides are integrals of $\hat{f}$ against the Gaussian transform. On that side multiplication by X_j is a derivative in ξ , and the Gaussian transform is its own derivative up to the factor $-(\Sigma\xi)_j$; the two integrands become equal. Nothing is inverted, so Σ may be singular.

(a) Put $\psi(\xi) = \mathbb{E}[e^{i\langle X, \xi \rangle}]$, which by Definition 1.3 of Lectures 1–2 equals $e^{-\frac{1}{2}\xi^\top \Sigma \xi}$ for every $\Sigma \succeq 0$. By Fourier inversion $f(x) = (2\pi)^{-d} \int \hat{f}(\xi) e^{i\langle x, \xi \rangle} d\xi$, and since $\mathbb{E} \int |\hat{f}(\xi) X_j| d\xi = \mathbb{E}|X_j| \|\hat{f}\|_{L^1} < \infty$, Fubini gives

$$\mathbb{E}[X_j f(X)] = \frac{1}{(2\pi)^d} \int \hat{f}(\xi) \mathbb{E}[X_j e^{i\langle X, \xi \rangle}] d\xi.$$

Differentiating under the expectation, $\partial_{\xi_j} \psi(\xi) = \mathbb{E}[i X_j e^{i\langle X, \xi \rangle}]$, while differentiating the formula for ψ gives $\partial_{\xi_j} \psi(\xi) = -(\Sigma\xi)_j \psi(\xi)$. Hence $\mathbb{E}[X_j e^{i\langle X, \xi \rangle}] = i(\Sigma\xi)_j \psi(\xi)$.

Since $\xi_k \hat{f} \in L^1$, the function $\partial_k f$ is recovered by inversion from $\widehat{\partial_k f} = i \xi_k \hat{f}$, and the same Fubini gives

$$\mathbb{E}[\partial_k f(X)] = \frac{1}{(2\pi)^d} \int i \xi_k \hat{f}(\xi) \psi(\xi) d\xi.$$

Multiply by Σ_{jk} and sum over k : the integrand becomes $i(\Sigma\xi)_j \hat{f}(\xi) \psi(\xi)$, which is the integrand of the first display. Only the formula $\psi(\xi) = e^{-\frac{1}{2}\xi^\top \Sigma \xi}$ was used; no square root or inverse of Σ appears, and the degenerate case needs no separate treatment.

(b) Fix v and take $F(x) = v f(\langle v, x \rangle)$ with $f \in C^1$ and f, f' bounded. Then $\langle W, F(W) \rangle = Y f(Y)$ for $Y = \langle v, W \rangle$, and $DF(x) = f'(\langle v, x \rangle) v v^\top$, so $\text{tr}(\Sigma DF) = \sigma^2 f'(Y)$ with $\sigma^2 = v^\top \Sigma v$. The hypothesis reads $\mathbb{E}[Y f(Y)] = \sigma^2 \mathbb{E}[f'(Y)]$. For $\sigma > 0$ this is Theorem 2.1 for Y/σ , so $Y \sim \mathcal{N}(0, \sigma^2)$; for $\sigma = 0$ take $f(y) = y/(1+y^2)$, which is admissible, to get $\mathbb{E}[Y^2/(1+Y^2)] = 0$ and hence $Y = 0$. Every linear functional of W is Gaussian, so $W \sim \mathcal{N}(0, \Sigma)$ by Corollary 1.7 of Lectures 1–2.

(c) Exercise 2.3(a) with $\{(k, l)\}$ as the index set: a matrix is a vector whose index is a pair, with covariance $\Sigma_{(ij), (kl)} = \mathbb{E}[M_{ij} M_{kl}]$. Nothing in that argument used the shape of the index set.

§2.5* Where independence was spent

2.13 A $\mathbb{E}F(W, W') = \mathbb{E}F(W', W)$ by exchangeability, and $= -\mathbb{E}F(W, W')$ by antisymmetry; so it is zero.

2.14 B $F(x, y) = (y - x)(f(x) + f(y))$ is antisymmetric, so $\mathbb{E}[(W' - W)(f(W) + f(W'))] = 0$. Writing $f(W') = f(W) + (W' - W)f'(W) + R$ with $|R| \leq \frac{\|f''\|_\infty}{2} (W' - W)^2$,

$$0 = 2\mathbb{E}[(W' - W)f(W)] + \mathbb{E}[(W' - W)^2 f'(W)] + \mathbb{E}[(W' - W)R].$$

The first term is $-2a\mathbb{E}[Wf(W)]$ by the Stein-pair property and conditioning on W . Rearranging gives the display, with the remainder bounded by $\frac{\|f''\|_\infty}{4a}\mathbb{E}|W' - W|^3$.

2.15 B Swapping $X_I \leftrightarrow X'_I$ is measure preserving, so (W, W') is exchangeable. Since $\mathbb{E}[X'_I | X_1, \dots, X_n] = 0$ and $\mathbb{E}[X_I | X_1, \dots, X_n] = \frac{1}{n} \sum_i X_i$, $\mathbb{E}[W' - W | X_1, \dots, X_n] = -\frac{1}{\sqrt{n}} \cdot \frac{1}{n} \sum_i X_i = -W/n$, and the tower property gives $a = 1/n$. Also $(W' - W)^2 = (X_I - X'_I)^2/n$, whose conditional expectation is $\frac{1}{n^2} \sum_i (X_i^2 + 1)$. Dividing by $2a = 2/n$ gives $\frac{1}{2}(1 + \frac{1}{n} \sum_i X_i^2)$, whose distance from 1 is $\frac{1}{2}|1 - \frac{1}{n} \sum_i X_i^2|$, at most $\frac{1}{2}\sqrt{\text{Var}(X_1^2)/n}$ in L^1 . The remainder is $\frac{n\|f''\|_\infty}{4}n^{-3/2}\mathbb{E}|X_1 - X'_1|^3$. Both are $O(n^{-1/2})$.

Where *independence entered*: in the two conditional expectations, and nowhere else. The product formula $\mathbb{E}[X_i f(W^{(i)})] = \mathbb{E}[X_i]\mathbb{E}[f(W^{(i)})]$ never appeared.

2.16 A $\mathbb{E}W = n \cdot \frac{1}{n} = 1$; $\mathbb{E}[I_i I_j] = \frac{1}{n(n-1)}$ for $i \neq j$ gives $\mathbb{E}W^2 = 1 + 1 = 2$, so $\text{Var } W = 1$. And $\mathbb{P}(I_1 = I_2 = 1) = \frac{1}{n(n-1)} \neq \frac{1}{n^2}$. What fails is **independence of the I_i** , which Theorem 2.10 requires; the theorem therefore says nothing, and a true conclusion reached from unsatisfied hypotheses is still not proved.

2.17 C

Force i to be a fixed point by **composing with the transposition that sends i to itself**: $\sigma = \pi \circ (i \ \pi^{-1}(i))$.

Then $\sigma(i) = \pi(\pi^{-1}(i)) = i$, and the map is a bijection from $\{\pi : \pi(i) \neq i\}$ onto itself composed with a relabelling, so σ is uniform among permutations fixing i . Writing $j = \pi^{-1}(i)$: i becomes fixed (gain 1), and j becomes fixed exactly when $\pi(i) = j$, i.e. when π has a 2-cycle at i (gain 2). Hence $V_i = W(\sigma) - 1 \in \{W, W + 1\}$, and

$$\mathbb{E}|W - V_i| \leq \mathbb{P}(\pi(i) = i) + \mathbb{P}(\text{2-cycle at } i) = \frac{1}{n} + \frac{1}{n}.$$

Chen–Stein with $p_i = 1/n$, $\lambda = 1$ gives $d_{\text{TV}}(W, \text{Poisson}(1)) \leq n \cdot \frac{1}{n} \cdot \frac{2}{n} = \frac{2}{n}$.

At $n = 10$ the bound is 0.2 and the truth is 2.3×10^{-8} : wrong by about seven orders of magnitude. What it has that inclusion–exclusion does not is **robustness** — it never used that π was a permutation beyond the coupling, and it is the only one of the three routes in this course that produces a bound at all.